

Hope and resilience: The unbroken thread of science

By Kartik Nemani

A quiet, powerful force has carried humanity through every epoch of uncertainty. Not technology, not policy, not even genius. It is *hope*, or the active choice to look toward a better future. And woven alongside it comes *resilience*, the stubborn refusal to stop.

Hope and resilience are perhaps the most underappreciated instruments in the scientist's toolkit. But together, they have guided civilizations through natural disasters, pandemics, and economic collapses.

The COVID-19 pandemic offers one of the most striking modern examples of these two forces' power. When conferences were canceled, collaborations severed, and laboratories shuttered for months, scientists did not simply pause—they improvised. Researchers found meaning in smaller, quieter victories: a paper finished, a new virtual collaboration, or a student mentored over a video call. When I look back now, what felt like interruption was, in many ways, a reinvention of our scientific processes to make them more adaptable to the realities of the modern world.

During the inaugural ACerS Spring Meeting in Bellevue, Wash., in April 2026, something quite remarkable happened. Tucked into the middle of busy technical symposia schedules was a roundtable discussion on the resilience of research in the United States. The session drew a full room, with attendees and panelists spanning the full range of scientific contributors, from students and early-career faculty to veteran academicians who have weathered decades of scientific funding cycles.

This kind of gathering matters more than it might seem. Science can be isolating, and the pressures of grant deadlines, publication records, and institutional metrics often make hope and

resilience feel like a private struggle. But moments such as these are what make science more intertwined and communal in spirit.

A poignant observation emerged from the discussion, one that resonated deeply with me. It was the reminder that we survived the dot-com collapse of 2001, the financial crisis of 2008, and the pandemic of 2020. So, we can survive the current deficit in federal funding for U.S. scientific research, too.

However, there is a difference between those past crises and the current uncertainty around federal funding for science. Those past events were not aimed at science itself, but the current situation strikes directly at the foundation of scientific research by questioning the value of fundamental studies.

The basic questions we ask at the lab bench today become the seeds for all applied innovations in the future. Consider the development of thermal barrier coatings. Basic research into the thermal and phase stability of yttria-stabilized zirconia in the 1970s laid the groundwork for the development of modern coatings, which today allow jet turbine blades to operate at temperatures 300°C above what the underlying metal could survive alone.

When society erodes the belief in basic science's value, we do not just delay scientific breakthroughs but sometimes compromise them entirely. As a result, there is genuine concern among today's researchers because while economic downturns correct themselves, mindset shifts can take generations to be undone.

And yet, hope is not lost because scientists are, by nature and training, adaptive. Those who lose their funding in one domain can find footing in another, emerging as phoenixes in adjacent fields. They carry their expertise into these new contexts, seeding disci-

plines they never originally intended to enter. This adaptability is not a failure of the system but an honest mirror to the intellectual versatility of the people who chose this path.

When I started pursuing a Ph.D., I found that the scientific community has always been more than its funding lines. It is a culture, a set of values, a commitment to truth-seeking that outlasts any particular political or economic moment. The gathering in Bellevue was proof of that. People did not fill that room because they had solutions but because they refused to be alone in this uncertainty.

Ultimately, the most durable source of hope in science is not policy reform or budget restoration, though both matter enormously. I believe it is the people who show up despite the uncertainty, from graduate students who find creative ways to run experiments despite the funding restrictions to professors who continue submitting proposals even after the previous ones were returned unfunded.

I strongly believe that resilience is not the absence of hardship but a perennial presence of purpose despite it. And as long as that purpose survives, as long as we continue to ask hard questions, mentor the next generation, and refuse to let curiosity be extinguished by circumstance, the very foundation—however shaken—will hold.

About the author

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